

SUBJECTIVE:

- * Patient presents with a 5-day history of very severe, sharp chest pain.
- * The pain is reported to radiate to the jaw.
- * The chest pain is aggravated by rest.
- * Patient also endorses an associated fever.

OBJECTIVE:

* Expected Physical Exam Findings:

- * General: Patient appears acutely ill, possibly diaphoretic and in significant distress.
- * Cardiovascular: Tachycardia expected. Auscultation may reveal a pericardial friction rub if pericarditis is present, or muffled heart sounds with a large effusion. No new murmurs, gallops (S3/S4) may be present.
- * Pulmonary: Lungs clear to auscultation bilaterally, without crackles or wheezes.
- * Abdomen: Soft, non-tender, non-distended, no hepatosplenomegaly.
- * Extremities: No edema, good capillary refill.
- * Vital Signs (Unavailable):
- * Temperature: Likely elevated (>100.4 F / 38 C).
- * Heart Rate: Tachycardic (>100 bpm).
- * Blood Pressure: Potentially elevated due to pain, or hypotensive if severe cardiac compromise.
- * Respiratory Rate: Tachypneic (>20 breaths/min).
- * Oxygen Saturation: Likely normal on room air.

ASSESSMENT:

- * Primary Diagnosis: Acute Pericarditis. The combination of sharp chest pain, fever, and severe pain is highly suggestive of pericardial inflammation.
- * Differential Diagnoses:
- * Acute Coronary Syndrome (ACS): Given very severe chest pain, sharp character, radiation to the jaw, and aggravation at rest, ACS must be a top concern and ruled out immediately.
- * Myocarditis: Often co-occurs with pericarditis or can present similarly with chest pain and fever, especially viral etiologies.
- * Aortic Dissection: Severe, sharp chest pain warrants consideration of this life-threatening condition, though fever is not typical.
- * Pulmonary Embolism (PE): While less typical with jaw radiation and sharp pain, severe chest pain requires consideration, especially if accompanied by dyspnea or hypoxemia.
- * Pleurisy/Pneumonia: Could cause sharp chest pain and fever, but jaw radiation is atypical.

PLAN:

* Tests:

- * Immediate 12-lead Electrocardiogram (ECG): Essential to evaluate for ST-segment changes, T-wave inversions, or diffuse ST elevation characteristic of pericarditis. Serial ECGs may be needed.
- * Cardiac Biomarkers: Troponin I/T, CK-MB to rule out myocardial injury/infarction.
- * Complete Blood Count (CBC) with differential, Erythrocyte Sedimentation Rate (ESR), C-Reactive Protein (CRP): To assess for inflammation and infection.
- * Chest X-ray (CXR): To evaluate for cardiomegaly, pulmonary congestion, or effusions.
- * Echocardiogram: To assess cardiac function, wall motion abnormalities, and presence/size of pericardial effusion.
- * Blood cultures: If infectious etiology for fever is strongly suspected.
- * Medications:
- * Pain control: Intravenous analgesia (e.g., Morphine) for severe pain.
- * Anti-inflammatory therapy: Once ACS is ruled out, high-dose NSAIDs (e.g., Ibuprofen) and Colchicine for pericarditis.
- * Aspirin: If ACS is suspected, administer immediately unless contraindicated.
- * Oxygen therapy: If oxygen saturation is $<92\%$.
- * Patient Advice:
- * Strict bed rest is recommended to reduce cardiac workload.
- * Report any worsening of pain, shortness of breath, or new symptoms immediately.
- * Avoid strenuous physical activity until cleared by cardiology.
- * Follow-up Instructions:

- * Admit to hospital for continuous cardiac monitoring, further diagnostic workup, and acute management.
- * Consult Cardiology immediately.